Immunocytochemical and immunohistochemical analysis

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Immunocytochemistry

Cells are detached from culturing flasks with trypsin, washed and collected in PBS, counted, and cytospinned on Polysine™ slides.

Cells are fixed in methanol-acetone (1:1) solution for 15 minutes at -20°C and washed in PBS.

Endogenous peroxidases are blocked by immersion of slides in distilled water with 3% H_2O_2 for 15 minutes.

Slides are washed with PBS.

Non-specific binding is inhibited by incubating the samples in 20% goat serum in PBS-0,05%Tween (PBS-T) for 1hr at 37°C in a humidified chamber.

Cells are incubated for 2 hr at room temperature in a humidified chamber with primary antibody (from rabbit or mouse) diluted in PBS-T. Negative controls are made with PBS-T alone.

Slides are washed with PBS twice.

Samples are incubated with biotinylated secondary goat antibody for 1 hr in a humidified chamber at room temperature (LSAB2 System-HRP kit, Dako).

Slides are washed with PBS twice.

Samples are incubated with HRP-streptavidin for 40 minutes in a humidified chamber at room temperature. (LSAB2 System-HRP kit, Dako).

Slides are washed with PBS twice.

The reaction is developed with DAB for a few minutes.

Slides are washed with distilled water and counterstained with Harris' hematoxylin for 3 seconds.

Slides are washed with running water for 10 minutes and then with distilled water.

Cells are dehydrated by immersion in 70% ethanol (5 min), in 96% ethanol (5 min), in 100% ethanol (2 times, 5 min) and in xylene (2 times, 7 min).

Cover slides are mounted with DPX.

Immunohistochemistry

Specimens are fixed for 24–48 hr in 10% neutral buffered formalin and paraffin embedded.

Specimens are deparaffinized by immersion in xylene 2 times for 30 min.

Specimens are rehydrated by immersion in 100% ethanol (2 times, 10 min.), in 96% ethanol (5 min), in 70% ethanol (5 min), in 50% ethanol (5 min) and in distilled water.

Antigen retrieval is performed by immersion in 0,01M citrate buffer (pH 6.0) in a microwave oven for 15 min at 750 W.

Samples are washed with PBS.

Endogenous peroxidases are blocked by immersion of slides in distilled water with 3% H_2O_2 for 15 minutes.

Samples are washed with PBS.

Non-specific binding is inhibited by incubating the samples in 20% goat serum in PBS-0,05%Tween (PBS-T) for 1hr in a humidified chamber at 37°C.

Specimens are incubated overnight at 4°C in a humidified chamber with primary antibody. Negative controls are made with PBS-T alone.

Slides are washed with PBS twice.

Samples are incubated with biotinylated secondary goat antibody for 1 hr in a humidified chamber at room temperature (LSAB2 System-HRP kit, Dako).

Slides are washed with PBS twice.

Samples are incubated with HRP-streptavidin for 40 minutes in a humidified chamber at room temperature. (LSAB2 System-HRP kit, Dako).

Slides are washed with PBS twice.

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